

DecodeLabs Artificial Intelligence Internship

Project 2 Report

Title

Data Classification Using Artificial Intelligence Using Logistic Regression

Submitted By

Name: Diksha Verma

Internship: DecodeLabs Artificial Intelligence Internship

Batch: 2026

1. Objective

The objective of this project is to build a machine learning classification model that can classify iris flowers into different species based on their physical characteristics. This project introduces the basic concepts of supervised machine learning, including data loading, preprocessing, model training, prediction, and evaluation.

2. Introduction

Artificial Intelligence enables computers to learn patterns from historical data and make predictions on new data. Classification is one of the most common supervised learning tasks where a model predicts the category or class of an input sample.

In this project, the famous Iris Flower Dataset is used to train a Logistic Regression model. The trained model learns the relationship between flower measurements and their corresponding species.

3. Dataset Description

The Iris dataset is a well-known dataset available in the Scikit-learn library.

It contains:

- Total Samples: 150
- Number of Features: 4
- Number of Classes: 3

Features

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

Target Classes

- Setosa
 - Versicolor
 - Virginica
-

4. Software and Libraries Used

Software

- Google Colab
- Python 3

Libraries

- Pandas
- NumPy

- Matplotlib
 - Scikit-learn
-

5. Methodology

The following steps were performed during the project:

Step 1

Imported all the required Python libraries.

Step 2

Loaded the Iris dataset from the Scikit-learn library.

Step 3

Converted the dataset into a Pandas DataFrame for better visualization and analysis.

Step 4

Checked the dataset shape, previewed the first few rows, and verified that there were no missing values.

Step 5

Split the dataset into training and testing sets using an 80:20 ratio.

Step 6

Created and trained a Logistic Regression classification model.

Step 7

Used the trained model to predict the species of the testing dataset.

Step 8

Evaluated the model using:

- Accuracy Score

- Classification Report
- Confusion Matrix

Step 9

Visualized the dataset using graphs.

6. Results

The Logistic Regression model successfully classified the iris flower species.

The following evaluation metrics were obtained:

- Accuracy Score
- Classification Report
- Confusion Matrix

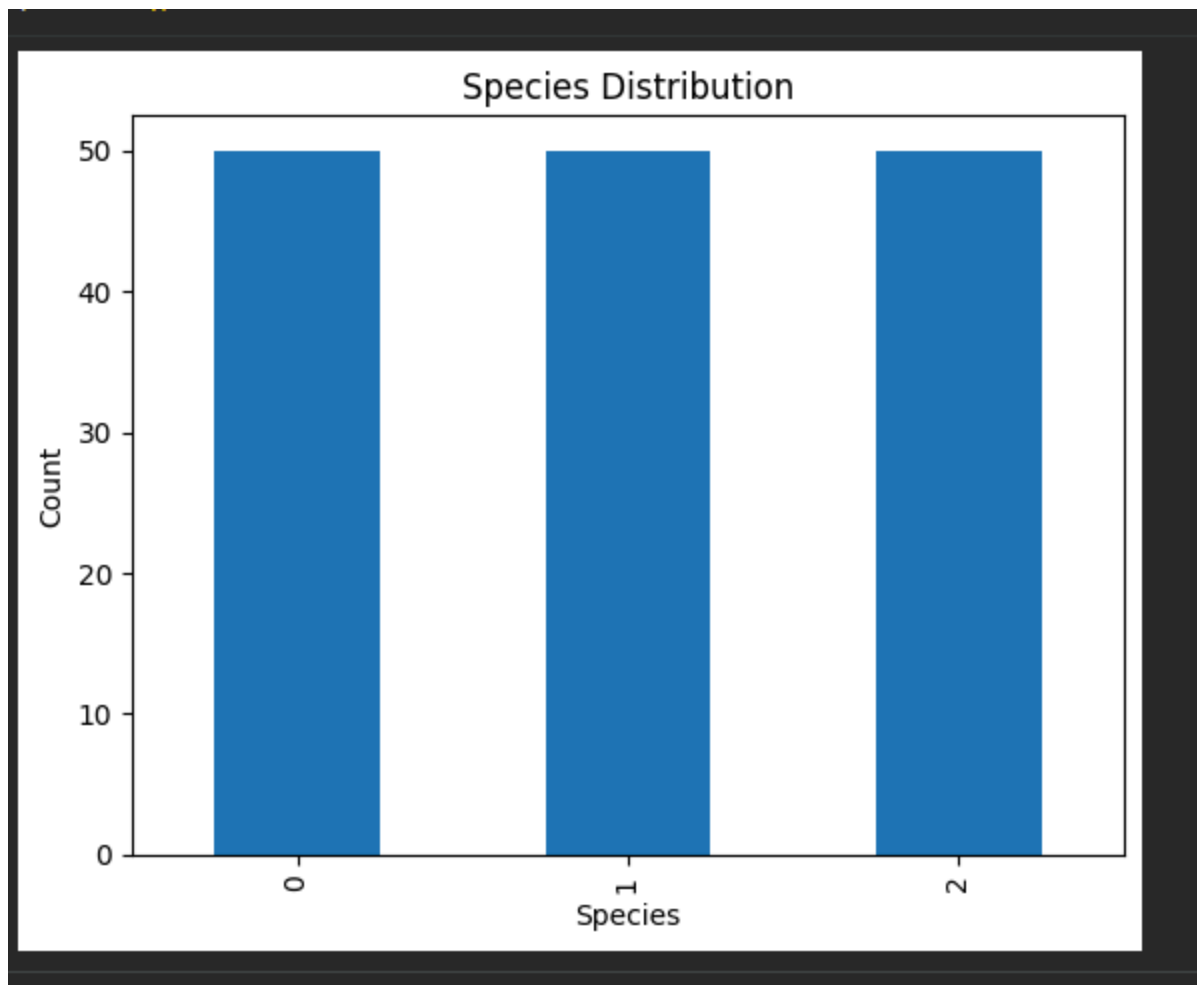
The model achieved a very high accuracy, demonstrating that Logistic Regression performs effectively on the Iris dataset.

7. Graphs

The following visualizations were created during the project:

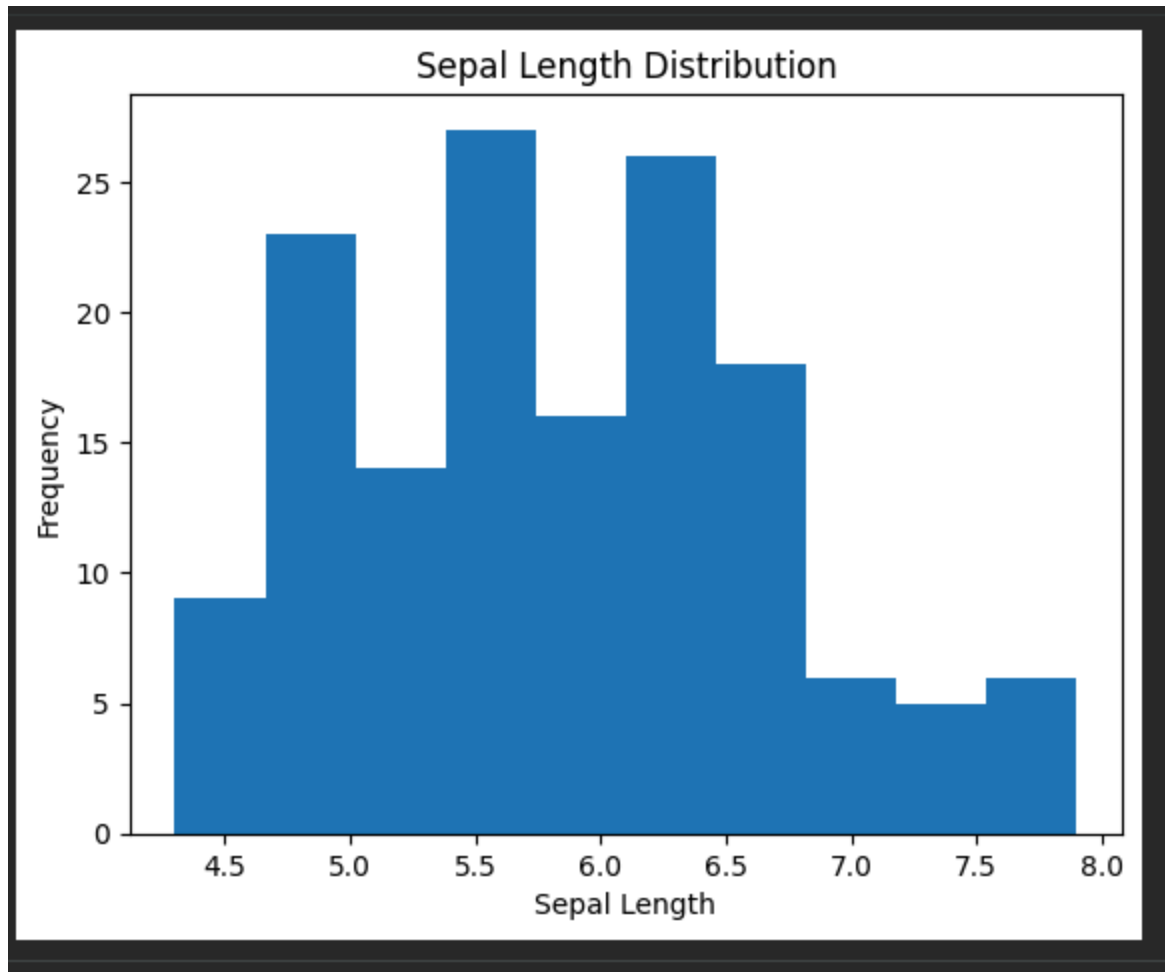
Graph 1

Species Distribution



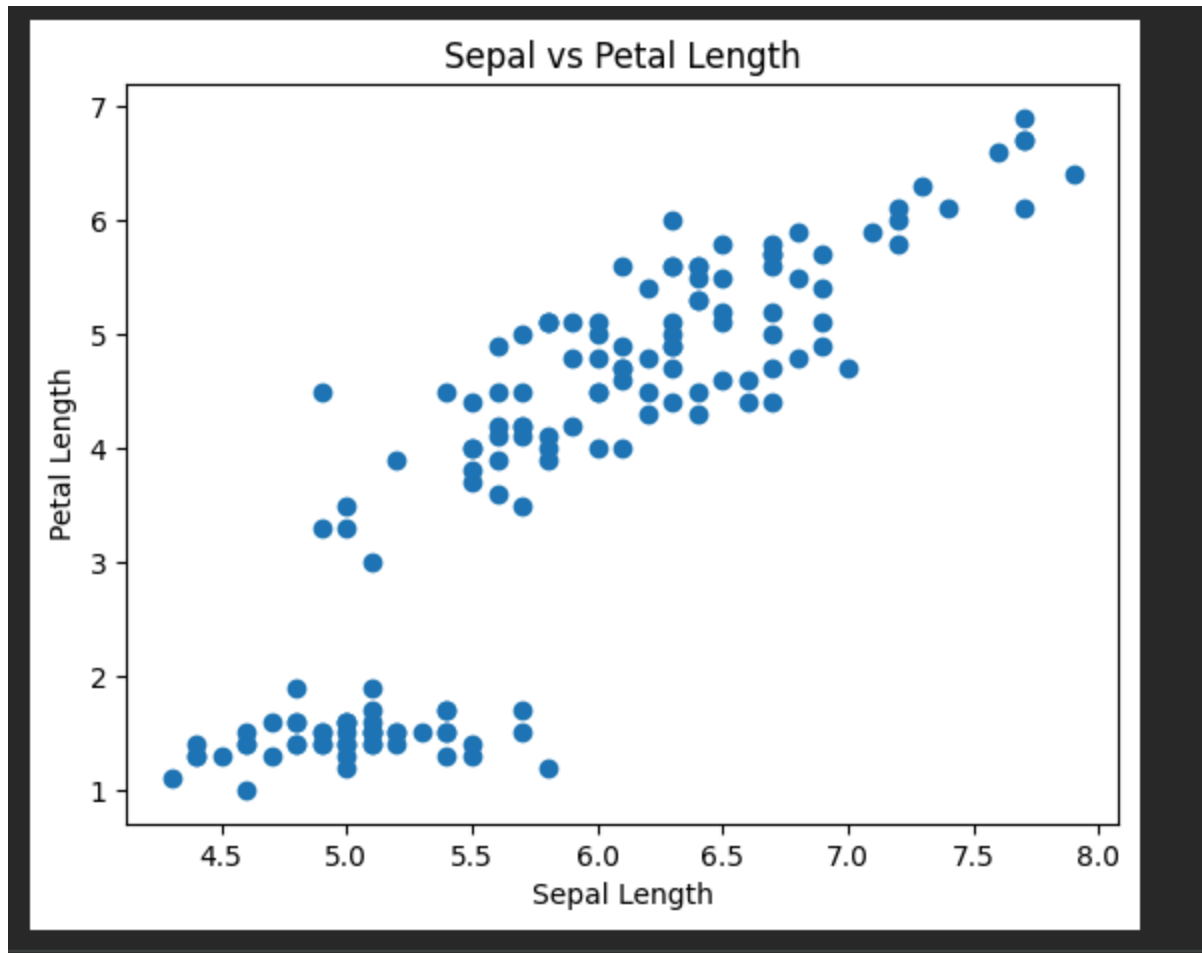
Graph 2

Sepal Length Distribution



Graph 3

Sepal Length vs Petal Length Scatter Plot



8. Advantages

- Simple and easy to understand.
 - Fast model training.
 - High classification accuracy.
 - Suitable for beginner machine learning projects.
 - Demonstrates the complete supervised learning workflow.
-

9. Applications

- Flower species identification
- Medical diagnosis

- Email spam detection
 - Customer segmentation
 - Image classification
 - Document categorization
-

10. Conclusion

This project successfully demonstrated the implementation of a supervised machine learning classification model using Logistic Regression. The Iris dataset was loaded, analyzed, and divided into training and testing datasets. The model was trained and evaluated using standard performance metrics. The achieved accuracy indicates that the model effectively learned the patterns in the data and accurately classified iris flower species. This project provides a strong foundation for understanding classification algorithms and supervised learning techniques in Artificial Intelligence.

11. Future Scope

The project can be further improved by:

- Using larger real-world datasets.
 - Comparing multiple classification algorithms such as Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine.
 - Performing hyperparameter tuning.
 - Deploying the trained model as a web application using Flask or Streamlit.
 - Exploring deep learning models for more complex classification tasks.
-

References

1. Scikit-learn Documentation
2. Python Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation

